6. [Maximum mark: 7]

Consider the quadratic polynomial given by $P(x) = 2x^2 + qx + r$ where $q, r \in \mathbb{R}$.

The equation $P(x) = 0$ has roots α and β, where $\alpha, \beta \in \mathbb{R}$.

When $P(x)$ is divided by $(x + 1)$, the remainder is 12.

Given that $\alpha^2 + \beta^2 = \frac{37}{4}$, find the value of q and the value of r.